Mathematics for Computer Science 1.1

Patrick Deng

June 2026

1. The Pythagorean Theorem says that if a and b are the lengths of the sides of a right triangle, and c is the length of the hypotenuse, then

$$a^2 + b^2 = c^2$$

This theorem is so fundamental and familiar that we generally take it for granted. But just being familiar doesn't justify calling it "obvious"—witness the fact that people have felt the need to devise different proofs of it for millennia. In this problem, we'll examine a particularly simple "proof without words" of the theorem. Here's the strategy. Suppose you are given four different colored copies of a right triangle with sides of lengths a, b, and c, along with a suitably sized square, as shown in Figure 1.1.

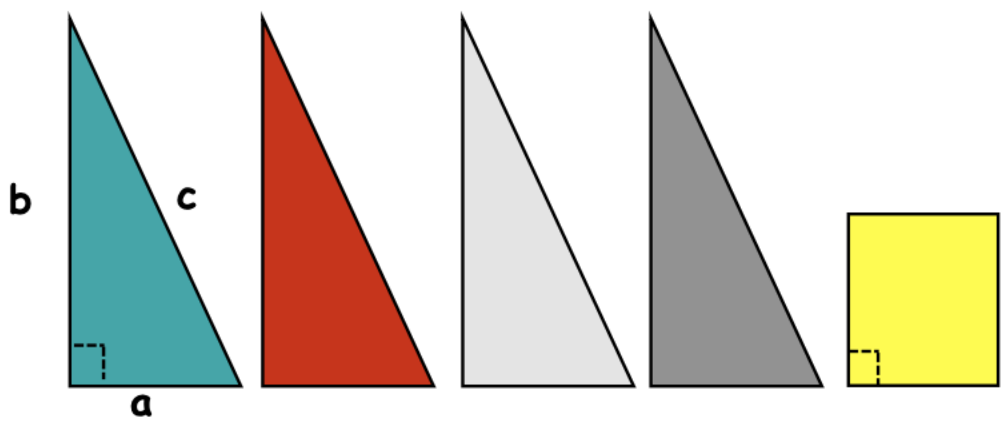
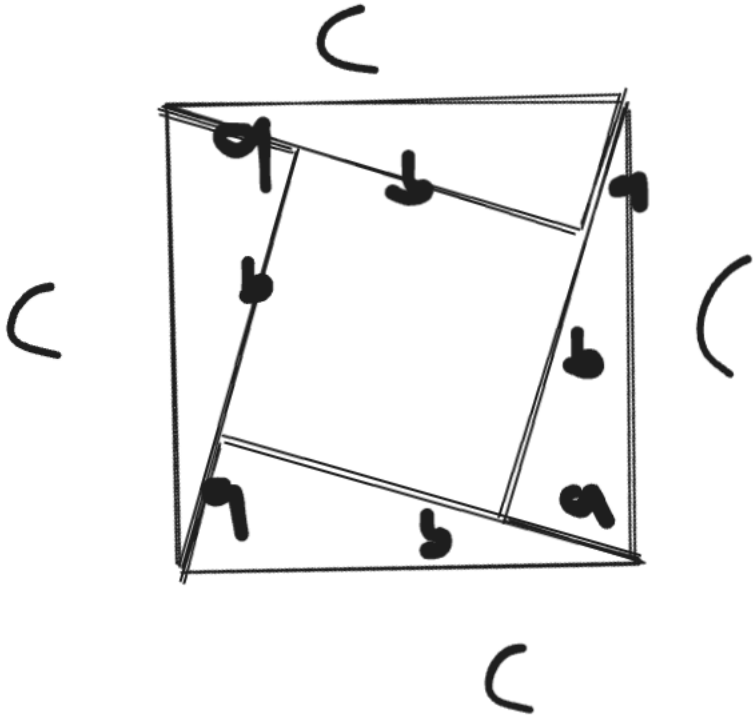


Figure 1.1 Right triangles and square.

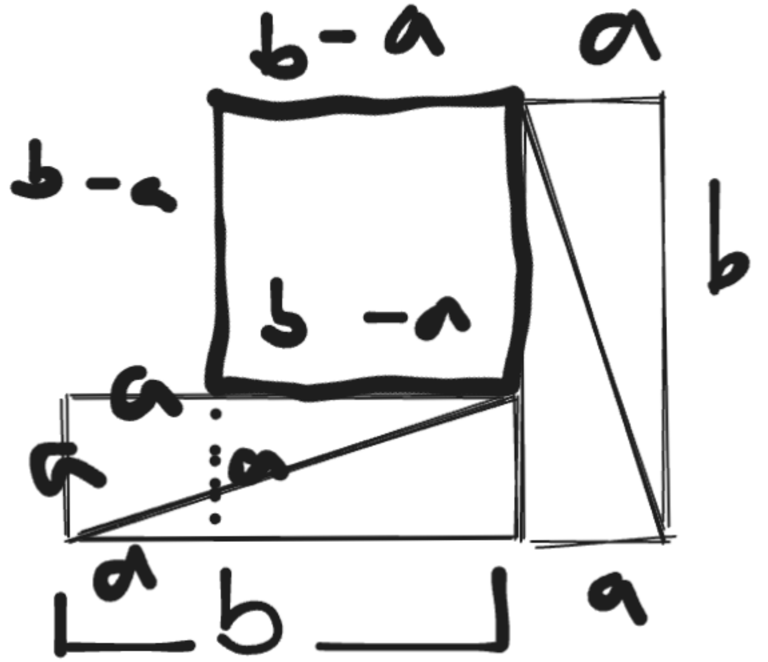
(a) You will first arrange the square and four triangles so they form a $c \times c$ square. From this arrangement you will see that the square is $(b - a) \times (b - a)$.



(b) You will then arrange the same shapes so they form two squares, one $a \times a$ and the other $b \times b$. You know that the area of an $s \times s$ square s^2. So appealing to the principle that

Area is preserved by Rearranging,

you can now conclude that $a^2 + b^2 = c^2$, as claimed. This is a really elegant and convincing proof of the Pythagorean Theorem, but it has some worrisome features. One concern is there might be something special about the shape of these particular triangles and square that makes the rearranging possiblefor example, suppose $a = b$?



(c) How would you respond to this concern?

All the side lengths would remain accurate no matter what, so it is okay to take the image as not to scale.

(d) Another concern is that a number of facts about right triangles, squares and lines are being *implicitly* assumed in justifying the rearrangements into squares. Enumerate some of these assumed facts.

If there is a segment $\overline{AB}$ and C is on this line segment, then $\overline{AC} + \overline{CB} = \overline{AB}$.

Area is preserved under rearrangement.

The total area of a shape is the sum of the areas of shapes it can be split into.

Rotating a shape and translating it does not alter its area.

2. What's going on here?!

$$1 = \sqrt{1} = \sqrt{(-1)(-1)} = \sqrt{-1}\sqrt{-1} = (\sqrt{-1})^2 = -1.$$

(a) Precisely identify and explain the mistake(s) in this *bogus* proof.

The statement $\sqrt{(-1)(-1)} = \sqrt{-1}\sqrt{-1}$ is false. It assumes that $\sqrt{ab} = \sqrt{a}\sqrt{b}$, but this does not hold for complex numbers, as $\sqrt{1} \neq \sqrt{-1}\sqrt{-1}$ even though $1 = (-1)(-1)$.

(b) Prove (correctly) that if $1 = -1$, then $2 = 1$.

If $1 = -1$ (given) and $1 = 1$ (already holds), $1 + 1 = -1 + 1$, so $2 = 0$. Since $2 = 0$, then $\frac{2}{2} = \frac{0}{2}$, so $1 = 0$. Finally, if $2 = 0$ and $1 = 0$, then $2 + 0 = 0 + 1$, so $2 = 1$.

(c) Every *positive* real number has two square roots, one positive and the other negative. The standard convention is that the expression $\sqrt{r}$ refers to the *positive* square root of r. Assuming familiar properties of multiplication of real numbers, prove that for positive real numbers r and s,

$$\sqrt{rs} = \sqrt{r}\sqrt{s}$$

First, we will prove that if arbitrary reals a, b are nonnegative and $a^2 = b^2$, then $a = b$.

We will prove this through the contraposition. If $a \neq b$, then $a^2 \neq b^2$. Without loss of generality, let $a < b$. We can then write $b = (a + (b - a))$, and if we square this we get:

$$(a + (b - a))^2 = a^2 + 2a(b - a) + (b - a)^2$$

This expression cannot be equal to a^2, as $2a(b - a) \geq 0$ and $(b - a)^2 > 0$ since $b - a > 0$. Thus, we have shown that $a, b \geq 0$ and $a^2 = b^2$ implies $a = b$. Now, we square $\sqrt{rs}$ and $\sqrt{r}\sqrt{s}$ to see if they yield the same number:

$$(\sqrt{rs})^2 = rs$$
$$(\sqrt{r}\sqrt{s})^2 = \sqrt{r}\sqrt{s} \cdot \sqrt{r}\sqrt{s} = rs$$

Thus, we conclude that $\sqrt{rs} = \sqrt{r}s$ for $r, s \geq 0$.

3. Identify exactly where the bugs are in each of the following bogus proofs.

(a) **Bogus Claim:** $\frac{1}{8} > \frac{1}{4}$

Bogus proof.

$$3 > 2$$
$$3\log_{10}(\frac{1}{2}) = 2\log_{10}(\frac{1}{2})$$
$$\log_{10}\left((\frac{1}{2})^3\right) > \log_{10}\left((\frac{1}{2})^2\right)$$
$$(\frac{1}{2})^3 > (\frac{1}{2})^2$$

It is not true that $3\log_{10}(\frac{1}{2}) > 2\log_{10}(\frac{1}{2})$, since $\log_{10}(\frac{1}{2}) < 0$, we must swap the $>$ into a $<$.

(b) *Bogus Proof:* 1¢= \$0.01 = (\$0.1)^2 = (10¢)^2 = 100¢= \$1.

Because a cent is actually a hundredth of a dollar, (10¢)^2 $\neq$ 100¢like the proof claims. (10¢)^2 = 1¢is true, on the other hand.

(c) **Bogus Claim:** If a and b are two equal real numbers, then $a = 0$.

Bogus proof.

$$a = b$$
$$a^2 = ab$$
$$a^2 - b^2 = ab - b^2$$
$$(a-b)(a+b) = (a-b)b$$
$$a + b = b$$
$$a = 0$$

The issue is the step that claims $a+b=b$. The proof successfully proves $(a-b)(a+b) = (a-b)b$ if $a = b$, but you cannot divide the equation by $a-b$ to get $a+b=b$, as this step only works assuming $a - b \neq 0$. Of course, since $a = b$, $a - b = 0$, so the proof is faulty.

4. It's a fact that the Arithmetic Mean is at least as large as the Geometric Mean, namely,

$$\frac{a+b}{2} \geq \sqrt{ab}$$

for all nonnegative real numbers a and b. But there's something objectionable about the following proof for the fact. What's the objection, and how would you fix it?

Bogus Proof.

$$\frac{a+b}{2} \overset{?}{\geq} \sqrt{ab} so$$
$$a + b \overset{?}{\geq} 2\sqrt{ab}$$
$$a^2 + 2ab + b^2 \overset{?}{\geq} 4ab$$
$$a^2 - 2ab + b^2 \overset{?}{\geq} 0$$
$$(a-b)^2 \geq 0$$

which we know is true.

The last statement is true because $a-b$ is a real number, and the square of a real number is never negative.

Though the steps are valid, we have not actually proved the original statement. What we have done is shown that if $\frac{a+b}{2} \geq \sqrt{ab}$, then $(a-b)^2 \geq 0$. However, both true statements and false statements can lead to a true statement, so this proof doesn't prove what it originally intended to.

This can be fixed by acknowledging all of the derived statements will have the same true value: Since the statements are all true or all false, and $(a-b)^2 \geq 0$ is certainly true, we know all statements are true, including $\frac{a+b}{2} \geq \sqrt{ab}$. Another way is to exchange the order. If we start out with a true statement $(a-b)^2 \geq 0$ and derive the desired statement $\frac{a+b}{2} \geq \sqrt{ab}$, we can conclude that it too must be true.

5. Albert announces to his class that he plans to surprise them with a quiz sometime next week.

   His students first wonder if the quiz could be on Friday of next week. They reason that it can't: if Albert didn't give the quiz before Friday, then by midnight Thursday, they would know the quiz had to be on Friday, and so the quiz wouldn't be a surprise any more.

   Next the students wonder whether Albert could give the surprise quiz Thursday. They observe that if the quiz wasn't given before Thursday, it would have to be given on the Thursday, since they already know it can't be given on Friday. But having figured that out, it wouldn't be a surprise if the quiz was on Thursday either. Similarly, the students reason that the quiz can't be on Wednesday, Tuesday, or Monday. Namely, it's impossible for Albert to give a surprise quiz next week. All the students now relax, having concluded that Albert must have been bluffing. And since no one expects the quiz, that's why, when Albert gives it on Tuesday next week, it really is a surprise!

   What, if anything, do you think is wrong with the students' reasoning?

   My personal take is that because the judge has sort of presented a paradox, it is entirely possible that Albert is not telling the truth. In other words, the students have based their conclusion on a false premise, so of course the conclusion he reaches is false. At the time of Albert's declaration, the students have no idea whether or not it is true.

   I don't actually have a clue though!

6. Show that $\log_7 n$ is either an integer or an irrational where n is a positive integer. Use whatever facts about integers and primes you need, but explicitly state such facts.

   Any number must either be rational or irrational. If it is rational number, then it can be represented as $\frac{n}{m}$, where n and m are coprime integers (meaning the $\gcd(n, m) = 1$ and $m > 0$. Furthermore, $\frac{n}{m}$ is an integer if $m = 1$ or $n = 0$. Thus, we must prove that if $\log_7 n$ can take the form $\frac{n}{m}$ with $n, m(\gcd(n, m) = 1)$ integers and $m > 0$, then $m > 1$ and $n \neq 0$.

   There are two cases, one if $\log_7 n$ is irrational, and the other for $\log_7 n$ being rational. If $\log_7 n$ is irrational, the case is trivial, as there is nothing to do. We now assume $\log_7 n$ to be rational.

$$\log_7 n = \frac{p}{q}$$

   where p, q are integers and $q > 0$. We will prove through contradiction that $q = 1$ must hold or $p = 0$. Assume that $p \neq 0$ and $q \neq 1$.

   Because $\log_7 n = \frac{p}{q}$, by the definition of the logarithm we have:

$$7^{\frac{p}{q}} = n$$

   Since $(x^{\frac{a}{b}})^b = x^a$ for real numbers x, a, b and $b \neq 0$, by exponentiating both sides by q we get

$$7^p = n^q$$

   There are two cases, $p > 0$ and $p < 0$. We first consider $p > 0$.

   Since $p > 0$ and p is an integer, 7^p is an integer too, as $7^p = 7 \cdot 7 \cdot \ldots 7$ p times for positive integer p. Thus, since $n^q = 7^p$, n^q is an integer too. We will use the fact that two integers are the same if and only if they have the same prime factorization. However, if n is a positive integer, it is impossible for n^q to have the same prime factorization as 7^p unless $\gcd(p,q) \neq 1$. We will prove this in the following paragraph:

We will prove this through contradiction. First, note that n's prime factorization must not contain any prime other than 7. If this is not the case, then that prime will be present in n^q's prime factorization, making it not equal to 7^p. Thus, $n = 7^k$, where k is a nonnegative integer. Substituting into the original equality, we have

$$(7^k)^q = 7^p$$

By exponent rules,

$$7^{kq} = 7^p$$

giving

$$kq = p$$

$$k = \frac{p}{q}$$

Recall that $\gcd(p,q) = 1$, $p \neq 0$ and $q \neq 1$. Note that k is not only a nonnegative integer, but a positive integer as $p \neq 0$. Since $\gcd(p,q) = 1$, $q = 1$ must hold, as if $q \neq 1$, then both n and q would be multiples of q, meaning $\gcd(p,q) \neq 1$. However, $q = 1$ cannot hold, so we have reached a contradiction! We now consider the case where $p < 0$.

$$7^p = n^q$$

Here, we see that n being a positive integer yields another contradiction, as n^q must be a positive integer (since q is a positive integer, $n^q = n \cdot n \cdot \dots \cdot n$ q times). However, 7^p is not an integer for negative p, as it is equivalent to $\frac{1}{7^{|p|}}$.

We have shown that assuming $p \neq 0$ and $q \neq 1$ leads to a contradiction through $7^p = n^q$. Thus, if $\log_7 n$ is rational and equal to $\frac{p}{q}$ for $q > 0$ and $\gcd(p,q) = 1$, then either $p = 0$ or $q = 1$, meaning $\frac{p}{q}$ must be an integer.

We finally conclude that $\log_7 n$ must either be irrational or an integer.

7. Prove by cases that

$$\max(r,s) + \min(r,s) = r + s$$

for all real numbers r, s.

There are 3 cases: $r > s$, $r = s$, and $r < s$.

Our first case is $r > s$.

$$\max(r,s) + \min(r,s) = r + s$$

If $r = s$, then

$$\max(r,s) + \min(r,s) = r + r = r + s$$

Lastly, for $r < s$

$$\max(r,s) + \min(r,s) = s + r = r + s$$

8. If we raise an irrational number to an irrational power, can the result be rational? Show that it can by considering $\sqrt{2}^{\sqrt{2}}$ and arguing by cases.

   We consider the following two cases: $\sqrt{2}^{\sqrt{2}}$ is rational, and $\sqrt{2}^{\sqrt{2}}$ is irrational. $\sqrt{2}^{\sqrt{2}}$ being rational is a trivial case, so we now consider $\sqrt{2}^{\sqrt{2}}$ being irrational.

   We are assuming that $\sqrt{2}^{\sqrt{2}}$ is irrational. Furthermore, we know $\sqrt{2}$ is irrational. We take $(\sqrt{2}^{\sqrt{2}})^{\sqrt{2}}$, which equals $\sqrt{2}^{(\sqrt{2}\cdot\sqrt{2})} = \sqrt{2}^2 = 2$, which is rational.

   Under both cases, we can find an irrational number raised to an irrational power being rational, so we have proved the desired statement.

9. Prove by cases that
$$|r + s| \leq |r| + |s|$$
   for all real numbers r, s.

   We now casework based on the signs of r and s.

   If $r \geq 0, s \geq 0$, then $|r + s| = r + s$ and $|r| + |s| = r + s$, so $|r + s| = |r| + |s|$, meaning $|r + s| \leq |r| + |s|$.

   Similarly, if $r < 0, s < 0$, then $|r+s| = -r-s$ and $|r|+|s| = -r-s$; once again we have $|r+s| \leq |r|+|s|$.

   We now handle the signs being different. Without loss of generality, let $r \geq 0$ and $s < 0$. $|r + s|$ must either equal to $r+s$ or $-r-s$. We also have $|r|+|s| = r-s$. If $|r+s| = r+s$, then $|r|+|s| = r-s \geq r+s$ since $-s \geq s$ $(s < 0)$. On the other hand, if $|r+s| = -r-s$, then $|r|+|s| = r-s \geq -r-s$ since $r \geq -r$ $(r \geq 0)$.

   Thus, for all cases we have $|r + s| \leq |r| + |s|$.

10. (a) Suppose that
$$a + b + c = d,$$
    where a, b, c, d are nonnegative integers.
    Let P be the assertion that d is even. Let W be the assertion that exactly one among a, b, c are even, and let T be the assertion that all three are even. Prove by cases that
$$P \text{ IFF } [W \text{ OR } T]$$

    We first show that P implies W OR T. If d is even, then of a, b, c, there must be an even number of odd numbers. Thus, there can either be 0 or 2 odd numbers. If there are zero odd numbers, then T is true. If there are two odd numbers, then W is true. Indeed, P implies W OR T.

    Now, we show W OR T implies P. We casework on whether W or T is true.

    If W is true, then exactly one of a, b, c is even. Without loss of generality, let a be even. Then, b, c are odd. Thus, $d = a + b + c$ must be even too, as $b + c$ is even and so is a.

If T is true, then all of a, b, c are even. Thus, $d = a + b + c$ is even too, so P holds.

We have proved both directions of the implication, so the statement is indeed true.

(b) Now suppose that

$$w^2 + x^2 + y^2 = z^2,$$

where w, x, y, z are nonnegative integers. Let P be the assertion that z is even, and let R be the assertion that all three of w, x, y are even. Prove by cases that

$$P \text{ IFF } R$$

Hint: an odd number equals $2m + 1$ for some integer m, so it's square equals $4(m^2 + m) + 1$.

It is clear that if w, x, y are all even, then $w^2 + x^2 + y^2$ will be a multiple of 4, as individually w^2, x^2, y^2 will be multiples of 4. Thus, if z is not even, then z^2 will not be a multiple of 4, making the equation impossible. Thus, it is true that R implies P.

We must now show that P implies R. We will show over all possible parities of w, x, y, that only the case where all are even numbers can satisfy the requirement that z is also even.

If none of w, x, y are even, then $w^2 + x^2 + y^2$ will be odd, thus z cannot be even, cause z^2 would be even too.

Let's consider the case where one of w, x, y is even. Without loss of generality, let w be even. We know that if $w^2 + x^2 + y^2$ is not a multiple of 4, then z cannot be even, giving us a contradiction. Since w^2 will be a multiple of 4, we wonder if $x^2 + y^2$ is a multiple of 4 where x, y are odd.

Let $x = 2k_1 + 1$ and $y = 2k_2 + 1$, where k_1, k_2 are integers. Then, $x^2 + y^2 = 4k_1^2 + 4k_1 + 1 + 4k_2^2 + 4k_2 + 1 = 4(k_1^2 + k_1 + k_2^2 + k_2) + 2$. Clearly, this number is not a multiple of 4, so we have a contradiction. Thus, $x^2 + y^2$ cannot be a multiple of 4 if x, y are odd, and $w^2 + x^2 + y^2$ also cannot be a multiple of 4. We conclude that if exactly one of w, x, y is even, then z cannot be even.

If two of w, x, y are even, then $w^2 + x^2 + y^2$ will be odd, so it is impossible for z to be even.

The final case is w, x, y all being even, which we already know means z must be even. Thus, we have shown that w, x, y being even implies z is even, and z can only be even if w, x, y are even, meaning P IFF R.

11. Prove that there is an irrational number a such that $a^{\sqrt{3}}$ is rational. *Hint*: Consider $\sqrt[3]{2}^{\sqrt{3}}$ and argue by cases.

We consider $\sqrt[3]{2}^{\sqrt{3}}$. We should first figure out whether $\sqrt[3]{2}$ is irrational or not. Let's assume $\sqrt[3]{2}$ is rational. Then,

$$\sqrt[3]{2} = \frac{p}{q}$$

for some coprime integers p, q with $q > 0$. Cubing both sides gives

$$2 = \frac{p^3}{q^3}$$

$$2q^3 = p^3$$

Since $2q^3$ is a multiple of 2, q^3 must be a multiple of 2, so q is also a multiple of 2. That would mean q^3 is actually a multiple of 8, so q has to be a multiple of 2 for $2q^3$to be a multiple of 2.

Since p, q are multiples of 2, they cannot be coprime. We have thus shown that $\sqrt[3]{2}$ is an irrational number, as if it were rational we find a contradiction.

Now, we will prove that an irrational number a must exist such that $a^{\sqrt{3}}$ is rational. We will casework on whether $\sqrt[3]{2}^{\sqrt{3}}$ is a rational or irrational number. If $\sqrt[3]{2}^{\sqrt{3}}$ is rational, then because $\sqrt[3]{2}$ is irrational, we can let $a = \sqrt[3]{2}$ be an irrational number such that $a^{\sqrt{3}}$ is rational.

If $\sqrt[3]{2}^{\sqrt{3}}$ is not rational, then we let $a = \sqrt[3]{2}^{\sqrt{3}}$. Then $a^{\sqrt{3}=\left(\sqrt[3]{2}^{\sqrt{3}}\right)^{\sqrt{3}}=\sqrt[3]{2}^{3}=2}$, which is rational.

Thus, no matter the case, we can find a valid a.

12. Prove that for any $n > 0$, if a^n is even, then a is even.
    Hint: Contradiction

    We will prove it through contradiction. Let a^n be even but a be odd. Then, a

13. Prove that if $a \cdot b = n$, then either a or b must be $\leq \sqrt{n}$, where a, b, and n are nonnegative real numbers.
    Hint: by contradiction, Section 1.8

    We will prove this through contradiction. Let $a > \sqrt{n}$ and $b > \sqrt{n}$. Then, $a = \sqrt{n} + \epsilon_1$ where $\epsilon_1 > 0$ and $b = \sqrt{n} + \epsilon_2$ where $\epsilon_2 > 0$. Multiplying a and b gives

$$(\sqrt{n} + \epsilon_1)(\sqrt{n} + \epsilon_2)$$

$$n + (\epsilon_1 + \epsilon_2)\sqrt{n} + \epsilon_1\epsilon_2$$

    Since $(\epsilon_1 + \epsilon_2)\sqrt{n} + \epsilon_1\epsilon_2$ is positive, $n + (\epsilon_1 + \epsilon_2)\sqrt{n} + \epsilon_1\epsilon_2$ must be greater than n.

14. Let n be a nonnegative integer.

    (a) Explain why if n^2 is even—that is, a multiple of 2—then n is even.

    If n is not a multiple of 2, so $n = (2k+1)$ for some integer k, then $n^2 = 4k^2 + 4k + 1$, which cannot be a multiple of 2. Thus, n must be a multiple of 2.

    (b) Explain why if n^2 is a multiple of 3, then n must be a multiple of 3.

    If n is not a multiple of 3, it takes the form of either $3k + 1$ or $3k + 2$.

    If $n = 3k + 1$, then $n^2 = 9k^2 + 6k + 1$, which is not a multiple of 3.

    If $n = 3k + 2$, then $n^2 = 9k^2 + 6k + 4 = 9k^2 + 6k + 3 + 1$, which also cannot be a multiple of 3. Thus, n must be a multiple of 3.

15. Give an example of two distinct positive integers m, n such that n^2 is a multiple of m, but n is not a multiple of m. How about having m be less than n?

    We can let $m = 4$ and $n = 2$. $n^2 = 4$ is a multiple of m, but n isn't.

    If we require $m < n$, then we could have $m = 4$ and $n = 6$. $n^2 = 36$ is a multiple of m, but n isn't.

16. How far can you generalize the proof of Theorem 1.8.1 that $\sqrt{2}$ is irrational? For example, how about $\sqrt{3}$?

    For the argument of $\sqrt{2}$ to work, we perform a proof by contradiction by assuming $\sqrt{n} = \frac{p}{q}$ where p, q are coprime integers and $q > 0$.

    We then get

    $$n = \frac{p^2}{q^2}$$

    $$nq^2 = p^2$$

    The goal is to show that p, q cannot be coprime. If n is not a perfect square, there is some prime number a such that a^k is in n's prime factorization and k is odd. We can thus conclude that p must contain a in its prime factorization as well, and the exponent of a in p's prime factorization must be at least $\frac{k+1}{2}$.

    We prove this through contradiction. If the exponent k_2 of a in p's prime factorization is less than $\frac{k+1}{2}$, we have $(p^{k_2})^2 = p^{2k_2}$. Since $k_2 < \frac{k+1}{2}$, $2k_2 < k + 1$. Since k is odd, $2k_2 < k$, since $2k_2$ cannot equal k, an odd number. We see that it is then impossible for $nq^2 = p^2$ to hold, as the exponent of a will not be correct.

    Thus, for any non perfect square n, $\sqrt{n}$ will be irrational. The square root of a perfect square must obviously be rational, so we have extended Theorem 1.8.1. If n is a positive integer and there does not exist an integer x such that $x^2 = n$, then $\sqrt{n}$ is irrational.

17. Prove that $\log_4 6$ is irrational.

    Let $n = \log_4 6$. We have

    $$4^n = 6$$

    We will do a proof by contradiction. Assume n is rational, that is, there exists integers p, q such that p and q are coprime and $q > 0$. Then,

    $$4^{\frac{p}{q}} = 6$$

    $$4^p = 6^q$$

    We can see that 6^q is a multiple of 3 as $q > 0$, but 4^p cannot be a multiple of 3. Thus, $\log_4 6$ must be irrational.

18. Prove by contradiction that $\sqrt{3} + \sqrt{2}$ is irrational.
    Hint: $(\sqrt{3} + \sqrt{2})(\sqrt{3} - \sqrt{2})$

    We will first prove that if x is rational and nonzero and y is irrational, then xy is irrational.

Since x is rational, $x = \frac{p}{q}$ for some coprime integers p, q with $q > 0$. Since x is not zero, $p \neq 0$. We will prove that xy is irrational through contradiction. Let $xy = \frac{m}{n}$ for some coprime integers m, n with $n > 0$.

$$xy = \frac{m}{n}$$
$$\frac{p}{q}y = \frac{m}{n}$$

Since $p \neq 0$, we can multiply both sides by $\frac{q}{p}$.

$$y = \frac{mq}{np}$$

Here, we see that y equals one integer mq divided by another integer np. This contradicts the fact that y is irrational, so we have thus proved that xy is irrational if x is a nonzero rational and y is irrational.

We will also prove that $x + y$ is irrational if x is rational and y is irrational, once again through contradiction. Let $x = \frac{p}{q}$ for coprime integers p, q with $q > 0$. We will assume $x + y$ is rational, meaning $x + y = \frac{m}{n}$ for coprime integers m, n with $n > 0$.

$$x + y = \frac{m}{n}$$
$$\frac{p}{q} + y = \frac{m}{n}$$
$$y = \frac{m}{n} - \frac{p}{q}$$
$$y = \frac{mq - np}{nq}$$

Here, we have shown that y is rational as it can be represented by one integer $(mq - np)$ divided by another nq. Thus, $x + y$ must be irrational if y is irrational and x is rational.

Assume that $\sqrt{3} + \sqrt{2} = x$, where x is a rational number. Then $x - 2\sqrt{2} = \sqrt{3} - \sqrt{2}$. Thus, we can multiply both sides of $\sqrt{3} + \sqrt{2} = x$ by this value.

$$(\sqrt{3} + \sqrt{2})(\sqrt{3} - \sqrt{2}) = x(x - 2\sqrt{2})$$
$$3 - 2 = x^2 - 2\sqrt{2}x$$
$$1 = x^2 - 2\sqrt{2}x$$

Since x is a rational number, x^2 is a rational number. $2\sqrt{2}x$ must be irrational, as $2x$ is rational and a nonzero rational number multiplied by an irrational is irrational as proved earlier. Since x^2 is rational and $2\sqrt{2}x$ is irrational, $x^2 + 2\sqrt{2}x$ must be irrational, as we proved earlier. Thus, $x^2 - 2\sqrt{2}x$ is irrational, and since $1 = x^2 - 2\sqrt{2}x$, 1 must be irrational too. However, this is clearly a contradiction.

Thus, we conclude that $\sqrt{3} + \sqrt{2}$ must be irrational through contradiction.

19. Here is a generalization of problem 1.16 which you may not have thought of:

Lemma. *Let the coefficients of the polynomial*

$$a_0 + a_1x + a_2x^2 + \cdots + a_{m-1}x^{m-1} + x^m$$

be integers. Then any real root of the polynomial is either integral or irrational.

(a) Explain why the Lemma immediately implies that $\sqrt[m]{k}$ is irrational whenever k is not an mth power of some integer.

Take $a_1 \dots a_{m-1} = 0$ and $a_0 = -k$. Then,

$$-k + x^m = 0$$

$$x^m = k$$

If k is not the mth power of an integer, then x cannot be an integer. However, because $x = \sqrt[m]{k}$ is a real root of the polynomial $-k + x^m$, by our lemma $\sqrt[m]{k}$ must be an irrational number if not an integer.

(b) Carefully prove the Lemma.
You may find it helpful to appeal to: **Fact.** If a prime p is a factor of some power of an integer, then it is a factor of that integer.

You may assume this Fact without writing down its own proof, but see if you can explain why it is true.

Let x be a root of the given polynomial. We must prove that x is either irrational or integral. To prove this contradiction, we assume $x = \frac{p}{q}$ where p, q are coprime integers but $q > 1$.

$$a_0 + a_1\frac{p}{q} + a_2\frac{p^2}{q^2} + \cdots + a_{m-1}\frac{p^{m-1}}{q^{m-1}} + \frac{p^m}{q^m} = 0$$

$$-\frac{p^m}{q^m} = a_0 + a_1\frac{p}{q} + a_2\frac{p^2}{q^2} + \cdots + a_{m-1}\frac{p^{m-1}}{q^{m-1}}$$

$$-p^m = a_0q^m + a_1pq^{m-1} + a_2p^2q^{m-2} + \cdots + a_{m-1}p^{m-1}q$$

$$-p^m = q(a_0q^{m-1} + a_1pq^{m-2} + a_2p^2q^{m-3} + \cdots + a_{m-1}p^{m-1})$$

We can see that the right hand side is a multiple of q, as $a_0, \dots, a_{m-1}$ are integers. Thus, $-p^m$ must also be a multiple of q. There must exist some prime number y that is a divisor of both $-p^m$ and q. For q to be a divisor of $-p^m$, all primes that are divisors of q are also divisors of $-p^m$. Thus, we can conclude that $-p^m$ and q are not coprime, as they share a prime factor.

As we have arrived at this contradiction, we have shown that x must either be an integer (so $q = 1$ and has no prime factor) or x is irrational.

20. Prove that $\log_9 12$ is irrational.

We will prove it through contradiction. Let $\log_9 12 = \frac{p}{q}$ for some coprime integers p, q with $q > 0$. We have

$$9^{\frac{p}{q}} = 12$$

$$9^p = 12^q$$

We can see that 9^p cannot be a multiple of 2, since it is the product of odd numbers, while 12^q must be even since it is the product of even numbers ($q > 0$). Thus, $\log_9 12$ must be irrational.

21. Prove that $\log_{12} 18$ is irrational.

We do yet another prove by contradiction. Let $\log_{12} 18 = \frac{p}{q}$ for some coprime integers p, q with $q > 0$. We have

$$12^{\frac{p}{q}} = 18$$

$$12^p = 18^q$$

$$2^{2p}3^p = 2^q 3^{2q}$$

We can see that this implies $2p = q$ and $p = 2q$, as if the exponents did not match up for either the 2 or the 3, the numbers' prime factorizations wouldn't be the same.

If $2p = q$, then substituting for q into $p = 2q$ gives $p = 4p$, so $p = 0$. Thus, $q = 0$ too. However, $q \neq 0$ must hold, so it is impossible for $\log_{12} 18$ to be rational.

22. A familiar proof that $\sqrt[3]{7^2}$ is irrational depends on the fact that a certain equation among those below is unsatisfiable by integers $a, b > 0$. (Note that more than one is unsatisfiable, but only one is relevant.) Indicate the equation that would appear in this proof, and explain why it is unsatisfiable. (Do *not* assume that $\sqrt[3]{7^2}$ is irrational.)

    i) $a^2 = 7^2 + b^2$
    ii) $a^3 = 7^2 + b^3$
    iii) $a^2 = 7^2 b^2$
    iv) $a^3 = 7^2 b^3$
    v) $a^3 = 7^3 b^3$
    vi) $(ab)^3 = 7^2$

    iv is the correct answer.

    $$a^3 = 7^2 b^3$$

    If the above statement is unsatisfiable by any integers $a, b > 0$, then neither is the statemeent

    $$\frac{a^3}{b^3} = 7^2$$

    $$\frac{a}{b} = \sqrt[3]{7^2}$$

    The final statement would imply that $\sqrt[3]{7^2}$ is rational, so if $a^3 = 7^2 b^3$ were unsatisfiable, then $\sqrt[3]{7^2}$ would be irrational.

    We now prove that the statement is unsatisfiable.

    Since $7^2 b^3$ is a multiple of 7, $a = 7^x y$, where x and y are both positive integers and y is not divisible by 7. Then, $a^3 = 7^{3x} y^3$. Let $b = 7^w z$, where w is a nonnegative integer and z is an integer not divisible by 7. This is certainly possible, as we are just pulling out the 7^w from b's prime factorization. We get

    $$7^{3x} y^3 = 7^2 (7^w z)^3$$

    $$7^{3x} y^3 = 7^{2+3w} z^3$$

    Since neither z^3 nor y^3 are multiples of 7, $3x = 2 + 3w$, otherwise the prime factorizations of both sides won't be the same. However, it is impossible for $3x = 3w + 2$, as $3w + 2$ is not a multiple of 3 while $3x$ is. Thus, We conclude that $a^3 = 7^2 b^3$ is not satisfiable for integers a, b and $\sqrt[3]{7^2}$ is irrational.

23. The fact that there are irrational numbers a, b such that a^b is rational was proved in Problem 1.8 by cases. Unfortunately, that proof was *nonconstructive*: it didn't reveal a specific pair a, b with this property. But in fact, it's easy to do this: let $a ::= \sqrt{2}$ and $b ::= 2\log_2 3$.
We know $a = \sqrt{2}$ is irrational, and $a^b = 3$ by definition. Finish the proof that these values a, b work, by showing that $2\log_2 3$ is irrational.

First, we note that, because the product of a rational number x and an irrational number y is irrational if $x \neq 0$, we only need to show $\log_2 3$ is irrational because 2 is rational and nonzero.

We will prove that $\log_2 3$ is irrational through contradiction. Let $\log_2 3 = \frac{p}{q}$ for some coprime integers p, q with $q > 0$. We have

$$2^{\frac{p}{q}} = 3$$

$$2^p = 3^q$$

We note that since $q > 0$, 3^q is a multiple of 3. However, 2^p cannot be a multiple of 3, so there are no p, q that satisfy $2^{\frac{p}{q}} = 3$.

We conclude that $\log_2 3$ is irrational, and as such, so is $2\log_2 3$.

24. Here is a different proof that $\sqrt{2}$ is irrational, taken from American Mathematical Monthly, v.116, 1, Jan. 2009, p.69:

Proof: Suppose for the sake of contradiction that $\sqrt{2}$ is rational, and choose the least integer $q > 0$ such that $(\sqrt{2} - 1)q$ is a nonnegative integer. Let $q' ::= (\sqrt{2} - 1)q$. Clearly $0 < q' < q$. But an easy computation shows that $(\sqrt{2} - 1)q'$ is a nonnegative integer, contradiction the minimality of q.

(a) This proof was written for an audience of college teachers, and at this point is a little more concise than desirable. Write out a more complete version which includes an explanation of each step.

Suppose for the sake of contradiction that $\sqrt{2}$ is rational, meaning $\sqrt{2} = \frac{m}{n}$ for some integers m, n with $n > 0$. Then,

$$\sqrt{2} - 1 = \frac{m}{n} - 1 = \frac{m - n}{n}$$

Clearly, $(\sqrt{2} - 1)n$ is an integer, so let q be the smallest integer > 0 such that $(\sqrt{2} - 1)q$ is a nonnegative integer. Note that $\frac{m-n}{n}$ must be nonnegative with $n > 0$ as $\sqrt{2} > 1$.

We let $q' = (\sqrt{2} - 1)q$. Since $\sqrt{2} - 1 < 1$ and $q > 0$, $0 < (\sqrt{2} - 1)q = q' < q$. However, recall that q' is also a nonnegative integer, and if we compute $(\sqrt{2} - 1)q'$ we get

$$(\sqrt{2} - 1)(\sqrt{2} - 1)q$$

$$(2 - 2\sqrt{2} + 1)q$$

$$(3 - 2\sqrt{2})q$$

$$((1 - \sqrt{2}) + (1 - \sqrt{2}) + 1)q$$

$$q - 2q'$$

Since q is an integer and so is q', we have shown that $(\sqrt{2} - 1)q'$ is also an integer. We are also confident that $(\sqrt{2} - 1)q'$ is nonnegative because q' is nonnegative and so is $\sqrt{2} - 1$.

So, we have shown that $(\sqrt{2}-1)q'$ is nonnegative and an integer, but $0 < q' < q$ and q was supposed to be the smallest nonzero integer such that $(\sqrt{2}-1)q$ is a nonnegative integer. We have achieved a contradiction, so we conclude that $\sqrt{2}$ cannot be rational.

(b) Now that you have justified the steps of this proof, do you have a preference for one of these proofs over the other? Why? Discuss these questions with your teammates for a few minutes and summarize your team's answers on your whiteboard.

I prefer the first way so long as it is possible for me to do the omitted steps myself. Thus, I can understand the main meat of the proof more easily. However, if parts of the proof which are not shown are unintuitive or nontrivial, I see that the flaws with the concise method may prevent more people from understanding the proof. Although more explanation can be helpful, I think that people think so differently that it is easier for a longer proof to be poorly written. Ultimately, I think it is up to the writer to write it for their audience as best as possible.

25. For $n = 40$, the value of polynomial $p(n) ::= n^2 + n + 41$ is not prime, as noted in Section 1.1. But we could have predicted based on general principles that no nonconstant polynomial can generate only prime numbers.
In particular, let $q(n)$ be a polynomial with integer coefficients, and let $c ::= q(0)$ be the constant term of q.

(a) Verify that $q(cm)$ is a multiple of c for all $m \in Z$.
We must show that $q(cm) = ck$ for some integer k. Let $q(n) = c + a_1 n + a_2 n^2 + \cdots + a_k n^k$, where a_i are all integers since we are told $q(n)$ has integer coefficients. Then, we have

$$q(cm) = c + a_1 c + a_2 c^2 + \cdots + a_k c^k$$

$$q(cm) = c(1 + a_1 + a_2 c + \cdots + a_k c^{k-1})$$

We can see that $q(cm)$ is indeed a multiple of c.

(b) Show that if q is nonconstant and $c > 1$, then as n ranges over the nonnegative integers N there are infinitely many $q(n) \in Z$ that are not primes.
Hint: You may assume that the magnitude of any nonconstant polynomial $q(n)$ grows as n grows.

From part a, we know $q(cm)$ is a multiple of c for integer m. We can take the polynomial $q'(m) = c + a_1 cm + a_2 c^2 m^2 + \cdots + a_k c^k m^k$, and since the magnitude of $q'(m)$ will increase unboundedly as m increases (as given by the hint), it must be the case that the magnitude of $q(cm) = q'(m)$ will also increase unbounded as m increases.

Thus, there are an infinite number of distinct $q(cm)$ which are multiples of c. The only multiple of c which could be prime is c, if c is prime. As such, there are also an infinite number of not-prime $q(cm)$.

(c) Conclude that for every nonconstant polynomial q there must be an $n \in N$ such that $q(n)$ is not prime. *Hint:* Only one case remains.

If the constant term c of the polynomial q is greater than 1, than we have already proved that an infinite number of $q(cm)$ exist such that $q(cm)$ is not prime.

It remains to consider $c = 0$ and $c = 1$. If $c = 0$, then $q(0) = 0$, which is not prime. If $c = 1$, then $q(1) = 1$, which is also not prime. Thus, for all possible values of c, there exists n such that $q(n)$ is not prime.